Bilal Amin

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Profile

Master's student in Data Science and AI with a background in mechatronics and hands-on experience spanning network simulation, data analysis, web development, and LLM applications. Interested in applied AI, intelligent software systems, and building data-driven products from prototype to deployment.

Experience

Research Assistant (HiWi) – Software Engineering Oct. 2026 – Present
Chair of Software Engineering, Saarland University *Saarbrücken, Germany*

- Work directly with Qingqing Dong on AI-assisted software engineering and model-driven system engineering research.
- Research focus includes investigating AI-assisted model merging, semantic merge-conflict detection, and feature-based merge recommendations.
- Support research on software evolution and intelligent methods for improving complex software and model-based systems.

Student Assistant (HiWi) – Web Development Sep. 2026 – Present
EuroComCenter, Saarland University *Saarbrücken, Germany*

- Maintain and improve a multilingual learning platform covering 6 Romance languages and structured around the 7-Sieves intercomprehension methodology.
- Reviewed and updated 10+ CMS-managed pages and content items, resolving technical, layout, and user-interface issues across the platform.
- Translate project requirements into practical improvements to navigation, content structure, usability, and multilingual presentation.

Working Student – Internet Architecture Group Oct. 2024 – Mar. 2025
Max Planck Institute for Informatics (MPI) *Saarbrücken, Germany*

- Designed an automated NS-3 evaluation framework comparing 5 congestion-control algorithms across 10 network scenarios each, covering 50 algorithm-scenario evaluations.
- Built a dashboard visualizing 4 key metrics – throughput, RTT/latency, packet loss, and congestion window – across experiment configurations.
- Developed 5 automated system tests and reusable evaluation logic to support repeatable performance experiments.
- Analyzed simulation results to identify algorithm-specific behavior and guide experiment and design decisions.

Data Analyst Oct. 2023 – Mar. 2024
Convergent Business Technologies *Islamabad, Pakistan*

- Completed structured training in programming, data science, and business analytics, applying C, Python, and SQL through hands-on assignments and mentor evaluations.
- Built 5 Tableau/Power BI dashboards and presented analytical findings to technical and business stakeholders in simulated client settings.
- Delivered an end-to-end Employee Attrition Prediction capstone using approximately 600 employee records, covering data preparation, modeling, evaluation, and presentation.
- Built a business-focused Dognition analytics dashboard and gained practical exposure to dimensional modeling, ETL concepts, SQL optimization, and Agile/Scrum workflows.

Web Development Intern Jun. 2023 – Aug. 2023
CureMD *Lahore, Pakistan*

- Built or significantly modified 18 responsive UI pages/components using HTML, CSS, and JavaScript for web-based healthcare software.
- Implemented approximately 6 .NET API endpoints to support application data exchange, backend integration, and internal tool development.

Technical Intern Aug. 2021 – Oct. 2021
STEP (Space Technology Education and Popularization) *Islamabad, Pakistan*

- Led 18 hands-on Scratch programming workshops – 6 larger and 12 smaller sessions – typically serving groups of 50–70 students.
- Created beginner-friendly educational material and practical activities to make programming and STEM concepts accessible to young learners.

Education

Universität des Saarlandes Apr. 2024 – Present
Master's degree, Data Science and AI – Grade: 1.9 *Saarbrücken, Germany*
University of Bergen – Erasmus+ Exchange Semester Jan. 2026 – Jun. 2026
Exchange Student, Computer Science *Bergen, Norway*
National University of Sciences & Technology Sep. 2019 – Sep. 2023
Bachelor's in Mechatronics Engineering (Computer Science) – Grade: 1.7 *Islamabad, Pakistan*

Projects

Musedots RFQ Agent – Atira EHL Grand Finale ↗ Oct. 2026
- Built a Python/Qwen/Playwright agent to investigate industrial tenders across four websites, match materials, calculate prices with Decimal, and validate and submit TAN-signed bids.
- Team musedots placed sixth in the private final evaluation with 100/100 quality and a 4:25 runtime; selected among the top seven teams for the finale pitch meeting.

AI-Assisted Dafny Code Generation and Verification ↗ 2026
- Built a Python/Codex pipeline that generates Dafny from English requirements and uses verifier feedback for up to two repairs.
- Compared two prompt profiles across three scenarios: all six runs verified within three attempts. Profiles tied in this small experiment; requirement coverage was reviewed separately from formal verification.

GermanLyft – AI-Powered German Learning Platform ↗ 2026
- Developed a full-stack German-learning platform for CEFR A1–A2 using React, TypeScript, and Flask, with placement tests, quizzes, accounts, profiles, and Anki `.apkg` importing.
- Built 4 exercise formats – multiple choice, fill-in-the-blank, sentence correction, and vocabulary matching – exposed through approximately 3 backend API endpoints.
- Tested the platform with 3 users and deployed the frontend on Vercel and backend on Render as 2 independently hosted services.

AI Learning Assistant using LLMs (RAG) 2025
- Built a local PDF learning assistant with Python, LangChain, Streamlit, FastAPI, Llama, and ChromaDB, offering citation-based answers, summaries, quizzes, and study materials.
- Tested retrieval across 25 Functional Programming documents (13 lectures and 12 worksheets), with 12+ questions per document and 300+ question-document interactions.

Additional projects: Smart StudyDesk ↗ (2025) – study organizer with usability testing; SoloScale (Side Hustle Launchpad) ↗ (2025) – FastAPI/MongoDB application at the Tech4Europe Hackathon, KIT; MarketPulse ↗ (2024) – responsive website and HCI analysis.

Technical Skills and Interests

Programming: Python, JavaScript, TypeScript, SQL, HTML, CSS, C++, C#, Java, Rust, Dafny
Machine Learning & AI: Scikit-Learn, PyTorch, TensorFlow, Hugging Face, LLMs, RAG, Prompt Engineering
LLM & Web Frameworks: LangChain, LangGraph, Ollama, ChromaDB, Codex SDK, FastAPI, Flask, Streamlit, React
Data & Tools: Pandas, NumPy, Matplotlib, Git, GitHub Actions, Playwright, Docker, Jupyter, Power BI, Tableau, Figma
Interests: Agentic AI, LLM Tool Calling, AI-Assisted Software Engineering, Browser and Workflow Automation

Selected Coursework and Certifications

Saarland University – MSc: Generative AI; Neural Networks; Elements of Machine Learning; Human-Computer Interaction; High-Level Computer Vision; Data Networks.

University of Bergen – Erasmus+: Functional Programming; Qualitative Analysis; Philosophy of AI.

Certifications: Machine Learning Specialization (DeepLearning.AI/Stanford); CS50 AI with Python (Harvard); Python for Data Science, AI & Development (IBM); Google Data Analytics Professional Certificate (in progress).